

# Dynamic Programming and Optimal Substructure

Dynamic programming solves complex problems by breaking them down into overlapping subproblems and storing intermediate solutions in a memoization table or 2D array.

Optimal substructure ensures the global optimum can be constructed from local subproblems.

# Dijkstra Shortest Path Algorithm

Finds the shortest paths between nodes in a graph with non-negative edge weights.

Uses a priority queue min-heap to achieve  $O(|E| + |V| \log |V|)$  asymptotic time complexity.